

Experiment 6

Develop Pig Latin scripts to sort, group, join, project, and filter the data.

1. Filter

```
101 John    CS    85
102 Alice   EC    92
103 Bob     CS    78
104 Eve     ME    88
105 Tom     CS    91
```

students.tsv

```
A = LOAD 'students.txt' AS (id:int, name:chararray, dept:chararray, marks:int);
B = FILTER A BY marks > 80;
DUMP B;
```

output

```
(101,John,CS,85)
(102,Alice,EC,92)
(104,Eve,ME,88)
(105,Tom,CS,91)
```

2. Project

```
A = LOAD 'students.txt' AS (id:int, name:chararray, dept:chararray, marks:int);
B = FOREACH A GENERATE name, marks;
DUMP B;
```

output

```
(John,85)
(Alice,92)
(Bob,78)
(Eve,88)
(Tom,91)
```

3. Join

```
101 John CS 85
102 Alice EC 92
103 Bob CS 78
104 Eve ME 88
105 Tom CS 91
```

students.tsv

```
CS Computer Science
EC Electronics
ME Mechanical
```

departments.tsv

```
A = LOAD 'students.txt' AS (id:int, name:chararray, dept:chararray, marks:int);
B = LOAD 'departments.txt' AS (code:chararray, deptname:chararray);
C = JOIN A BY dept, B BY code;
DUMP C;
```

```
(101,John,CS,85,CS,Computer Science)
(102,Alice,EC,92,EC,Electronics)
(103,Bob,CS,78,CS,Computer Science)
(104,Eve,ME,88,ME,Mechanical)
(105,Tom,CS,91,CS,Computer Science)
```

output

4. Group

```
A = LOAD 'students.txt' AS (id:int, name:chararray, dept:chararray, marks:int);  
B = GROUP A BY dept;  
DUMP B;
```

```
(CS, {(101, John, CS, 85), (103, Bob, CS, 78), (105, Tom, CS, 91)})  
(EC, {(102, Alice, EC, 92)})  
(ME, {(104, Eve, ME, 88)})
```

output

5. Sort

```
A = LOAD 'students.txt' AS (id:int, name:chararray, dept:chararray, marks:int);  
B = ORDER A BY marks DESC;  
DUMP B;
```

```
(102, Alice, EC, 92)  
(105, Tom, CS, 91)  
(104, Eve, ME, 88)  
(101, John, CS, 85)  
(103, Bob, CS, 78)
```

output